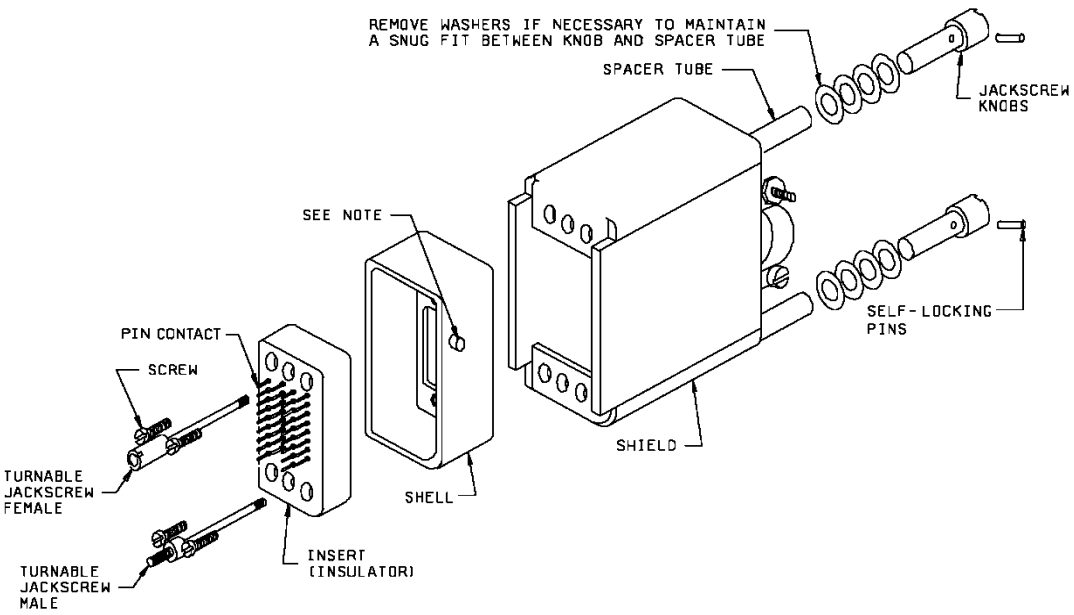
INCH-POUND
MIL-DTL-28748/3F
w/AMENDMENT 2
9 January 2017
SUPERSEDING
MIL-DTL-28748/3F
w/AMENDMENT 1
8 August 2014

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RACK AND PANEL, CRIMP TYPE REMOVABLE PIN CONTACTS, SIZES 16 AND 20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28748.



Connector

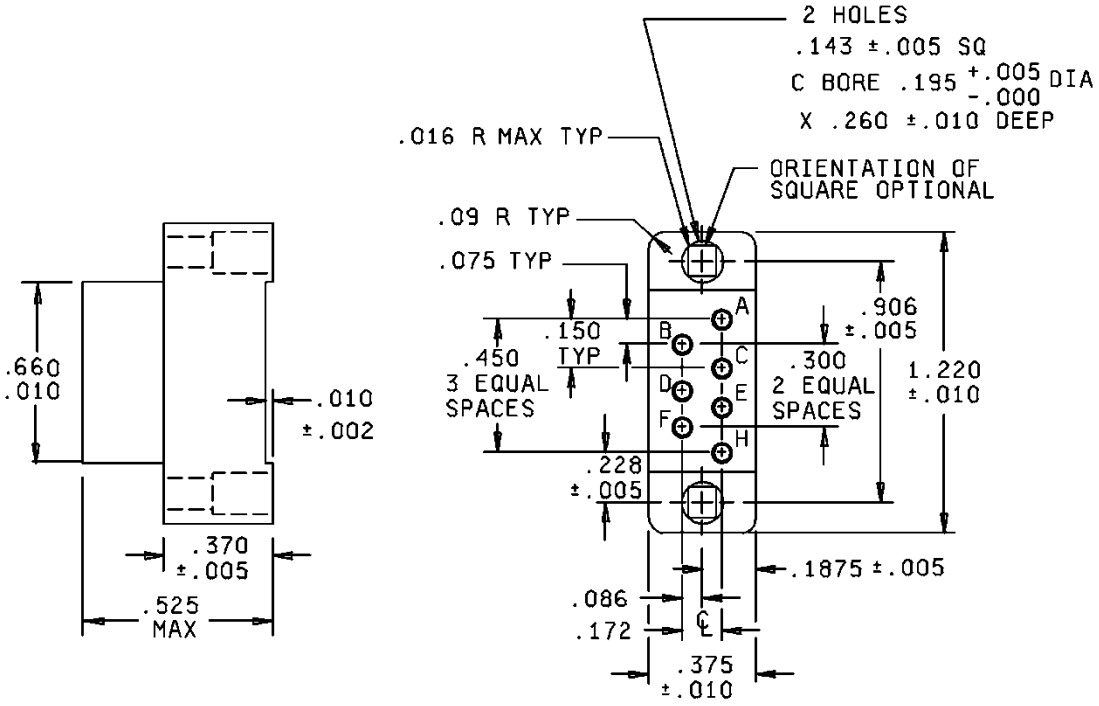
NOTE: The polarization feature (pin on the plug shell, groove on the receptacle shell) shall be adjacent to the side closest to the number 1 or A contact position.

Configuration A

FIGURE 1. Plug connector with shield, plug shell, and turnable jackscrews.

AMSC N/A FSC 5935

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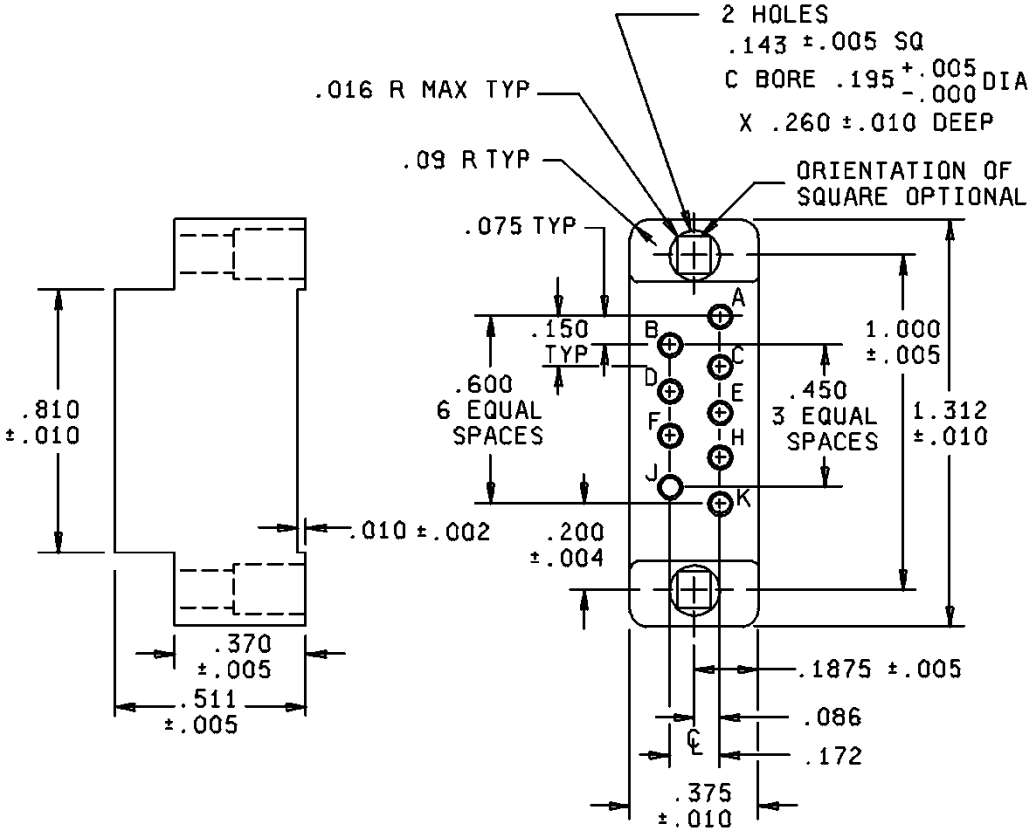
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.143	3.63	.300	7.62
.005	0.13	.150	3.81	.370	9.40
.010	0.25	.172	4.37	.375	9.53
.016	0.41	.1875	4.765	.450	11.43
.075	1.91	.195	4.95	.525	13.34
.086	2.18	.228	5.79	.660	16.76
.09	2.3	.260	6.60	.906	23.01
				1.220	30.99

Insert with 7 contacts

Insert designator A

FIGURE 2. Insert arrangements.

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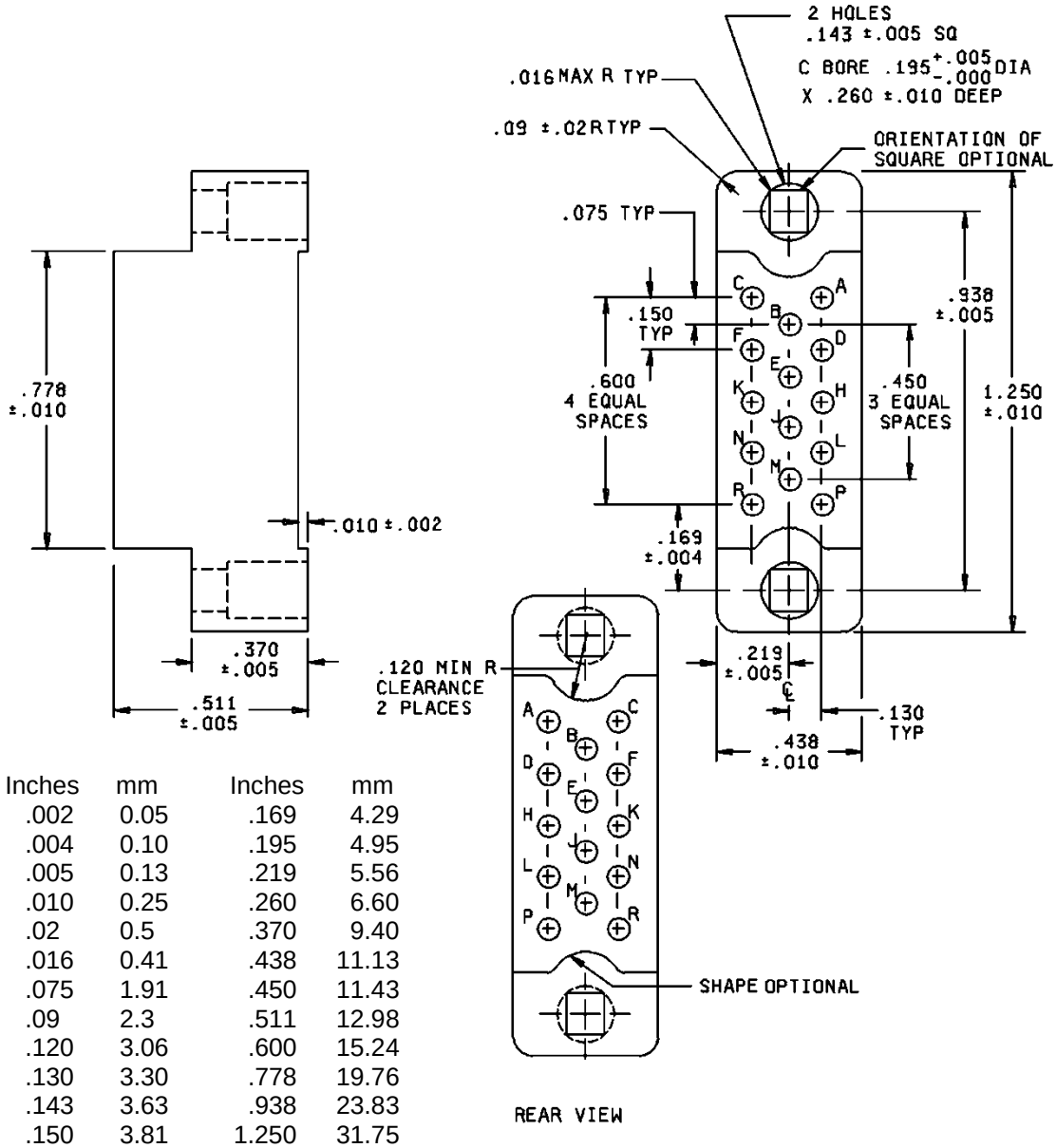
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.143	3.63	.375	9.53
.004	0.10	.150	3.81	.450	11.43
.005	0.13	.172	4.37	.511	12.98
.010	0.25	.1875	4.76	.600	15.24
.016	0.41	.195	4.95	.810	20.57
.075	1.91	.200	5.08	1.000	25.40
.086	2.18	.260	6.60	1.312	33.32
.09	2.3	.370	9.40		

Insert with 9 contacts

Insert designator B

FIGURE 2. Insert arrangements - Continued.

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Check the source to verify that this is the current version before use.



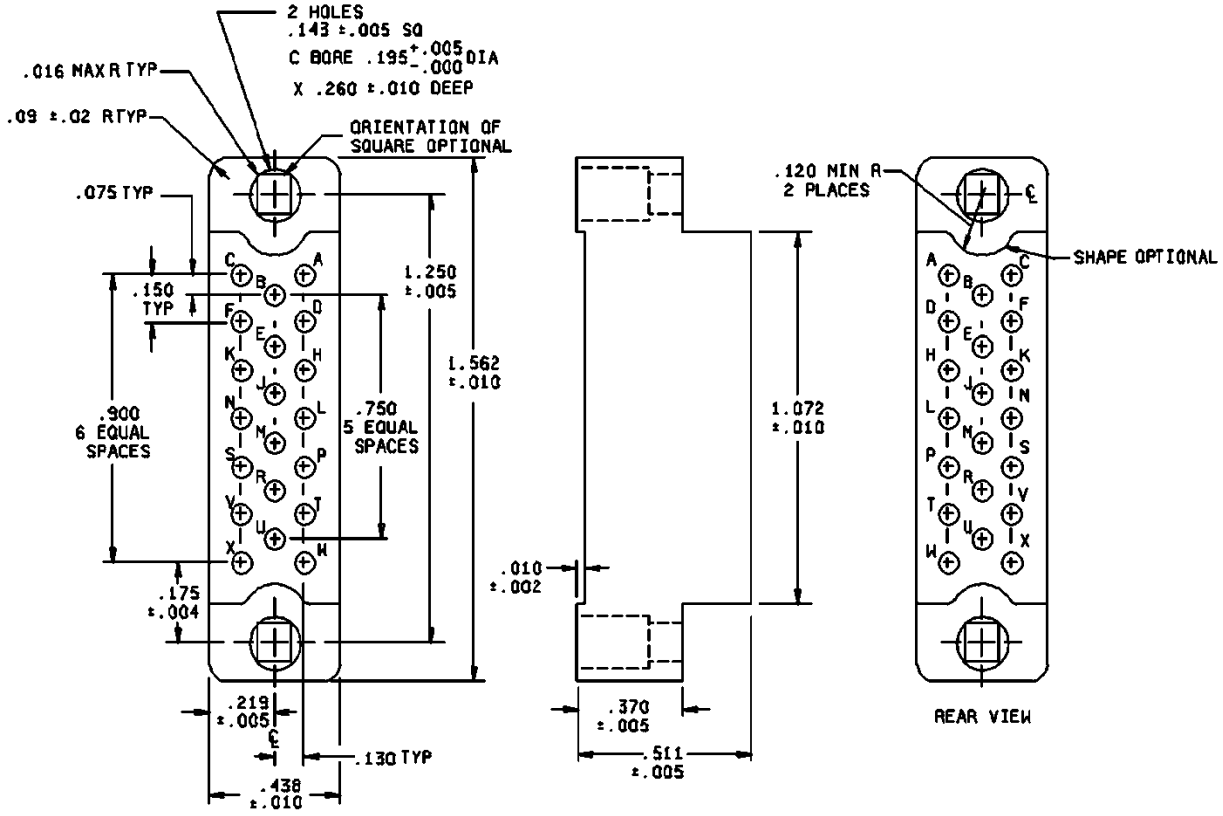
Inches	mm	Inches	mm
.002	0.05	.169	4.29
.004	0.10	.195	4.95
.005	0.13	.219	5.56
.010	0.25	.260	6.60
.02	0.5	.370	9.40
.016	0.41	.438	11.13
.075	1.91	.450	11.43
.09	2.3	.511	12.98
.120	3.06	.600	15.24
.130	3.30	.778	19.76
.143	3.63	.938	23.83
.150	3.81	1.250	31.75

Insert with 14 contacts

Insert designator C

FIGURE 2. Insert arrangements - Continued.

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Check the source to verify that this is the current version before use.



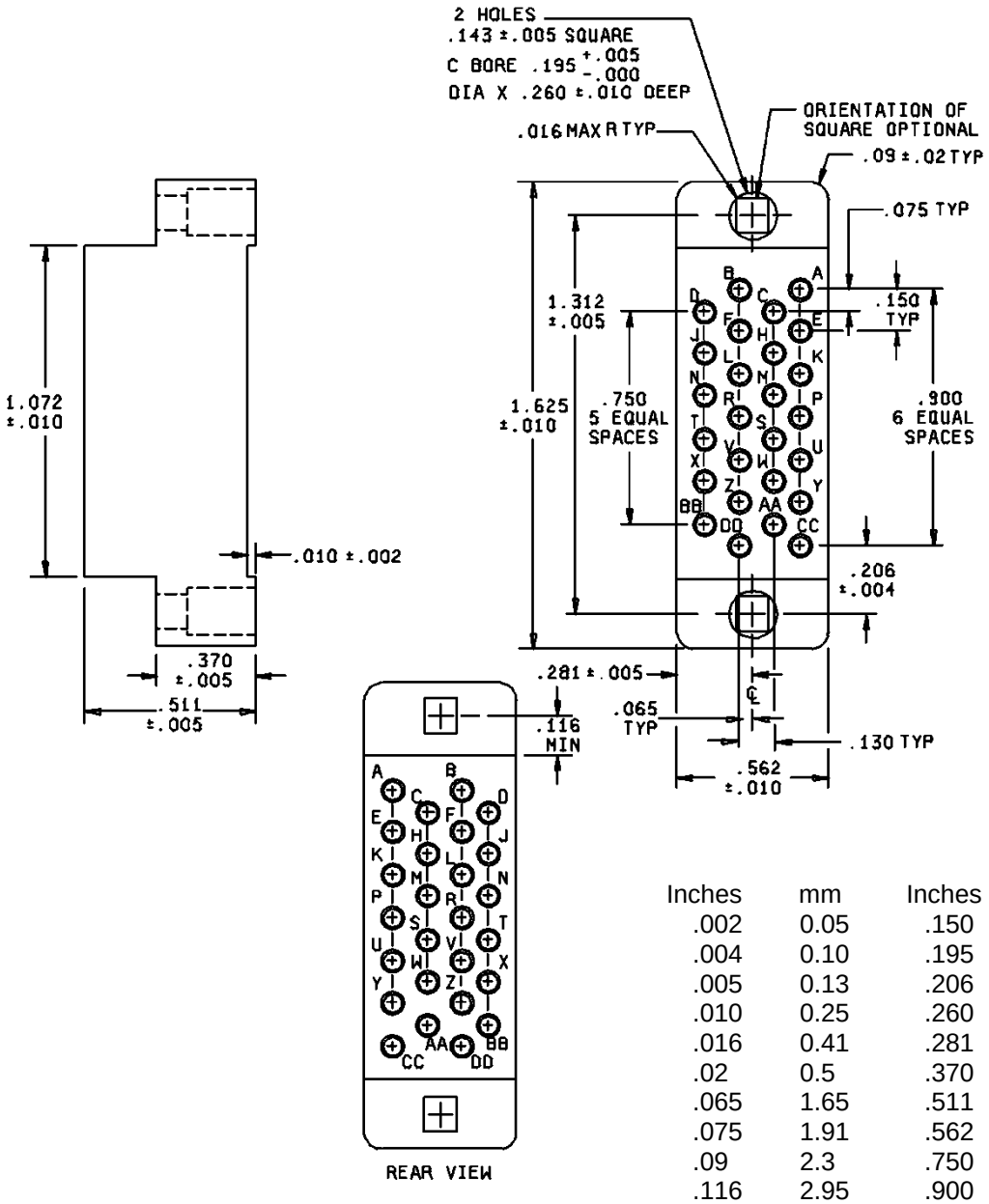
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.120	3.05	.370	9.40
.004	0.10	.130	3.30	.438	11.13
.005	0.13	.143	3.63	.511	12.98
.010	0.25	.150	3.81	.750	19.05
.016	0.41	.175	4.45	.900	22.86
.02	0.5	.195	4.95	1.072	27.23
.075	1.91	.219	5.56	1.250	31.75
.09	2.3	.260	6.60	1.562	39.67
				2.290	58.17

Insert with 20 contacts

Insert designator D

FIGURE 2. Insert arrangements - Continued.

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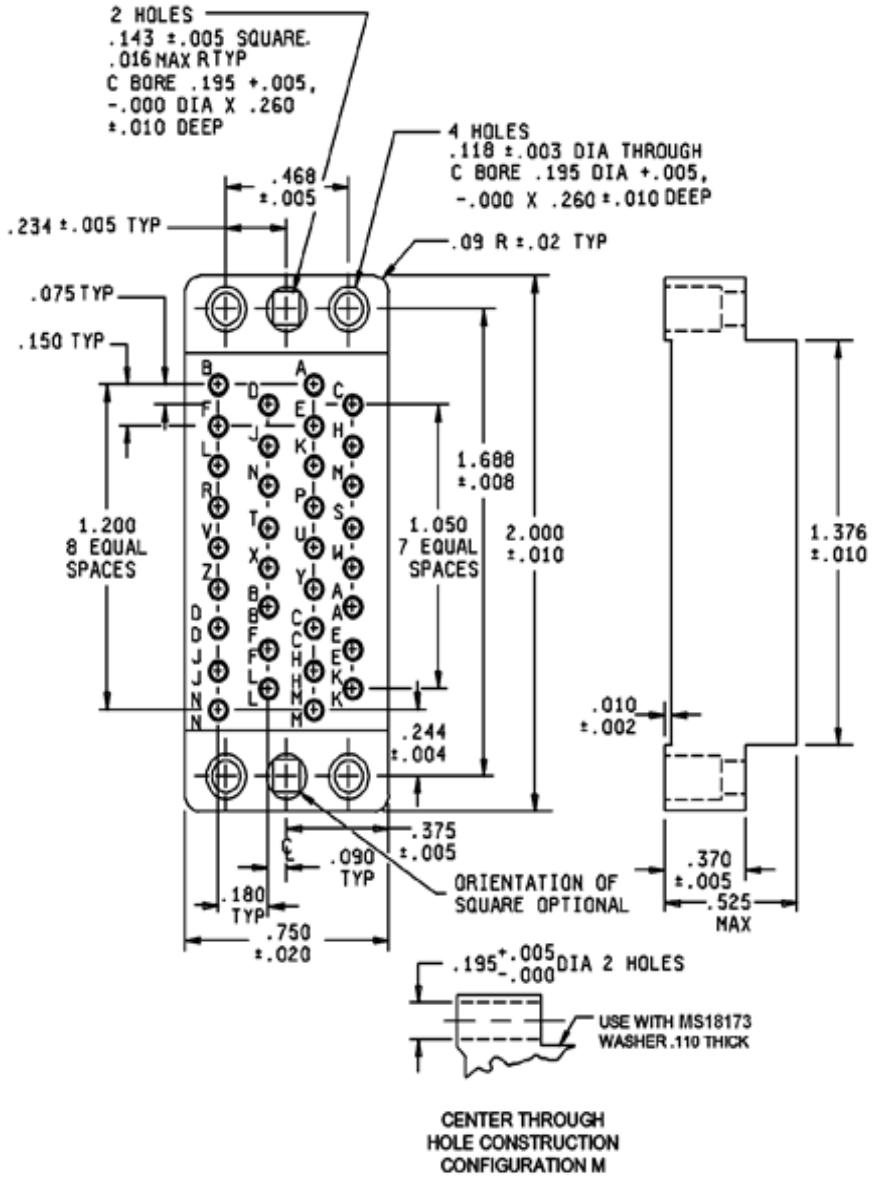
Inches	mm	Inches	mm
.002	0.05	.150	3.81
.004	0.10	.195	4.95
.005	0.13	.206	5.23
.010	0.25	.260	6.60
.016	0.41	.281	7.14
.02	0.5	.370	9.40
.065	1.65	.511	12.98
.075	1.91	.562	14.28
.09	2.3	.750	19.05
.116	2.95	.900	22.86
.143	3.63	1.072	27.23
.130	3.30	1.312	33.32
.143	3.63	1.625	41.28

Insert with 26 contacts

Insert designator E

FIGURE 2. Insert arrangements - Continued.

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Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.004	0.10
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.02	0.5
.020	.051
.075	1.91
.09	2.3
.090	2.29
.110	2.79
.118	3.00
.143	3.63
.150	3.81
.180	4.57
.195	4.95
.234	5.94
.244	6.20
.260	6.60
.370	9.40
.375	9.53
.468	11.88
.525	13.34
.750	19.05
1.050	26.67
1.200	30.48
1.376	34.95
1.688	42.88
2.000	50.80

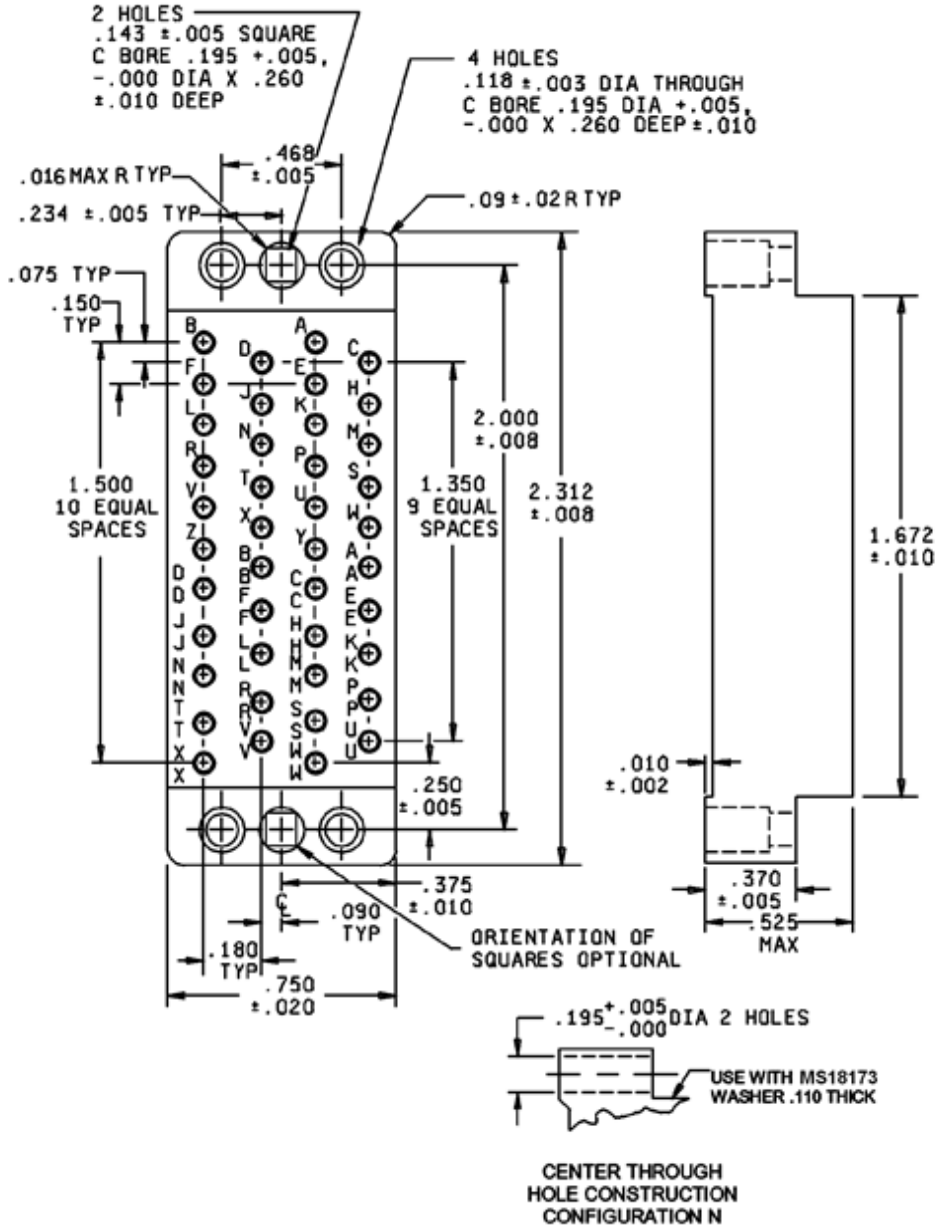
Configuration F - Suitable for all applications, except for one-piece jackscrews.
Configuration M - For long jackscrews and shield only (through hole alternate construction).

Insert with 34 contacts

Insert designators F and M

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.02	0.5
.020	0.51
.075	1.91
.09	2.3
.090	2.29
.110	2.79
.118	3.00
.143	3.63
.150	3.81
.180	4.57
.195	4.95
.234	5.94
.250	6.35
.260	6.60
.370	9.40
.375	9.53
.468	11.89
.525	13.34
.750	19.05
1.350	34.29
1.500	38.10
1.672	42.47
2.000	50.80
2.312	58.72

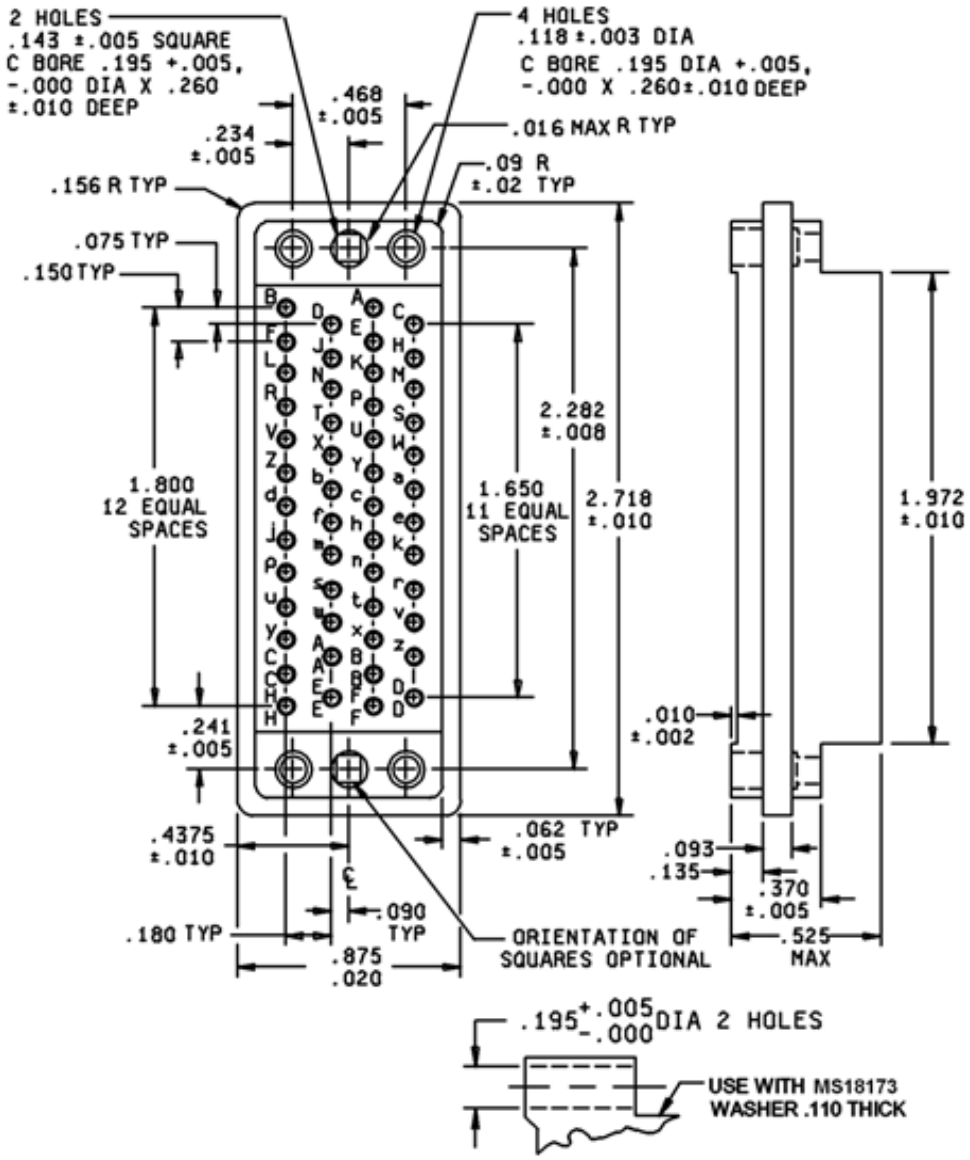
Configuration G - Suitable for all applications, except one-piece jackscrews.
Configuration N - For long jackscrews and shield only (through hole alternate construction).

Insert with 42 contacts

Insert designators G and N

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
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CENTER THROUGH
HOLE CONSTRUCTION
CONFIGURATION P

Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.02	0.5
.020	0.51
.062	1.57
.075	1.91
.09	2.3
.090	2.29
.093	2.36
.110	2.79
.118	3.00
.135	3.43
.143	3.63
.150	3.81
.156	3.96
.180	4.57
.195	4.95
.234	5.94
.241	6.12
.260	6.60
.370	9.40
.4375	11.11
.468	11.89
.525	12.98
.875	22.23
1.650	41.91
1.800	45.72
1.972	50.09
2.282	57.96
2.718	69.04

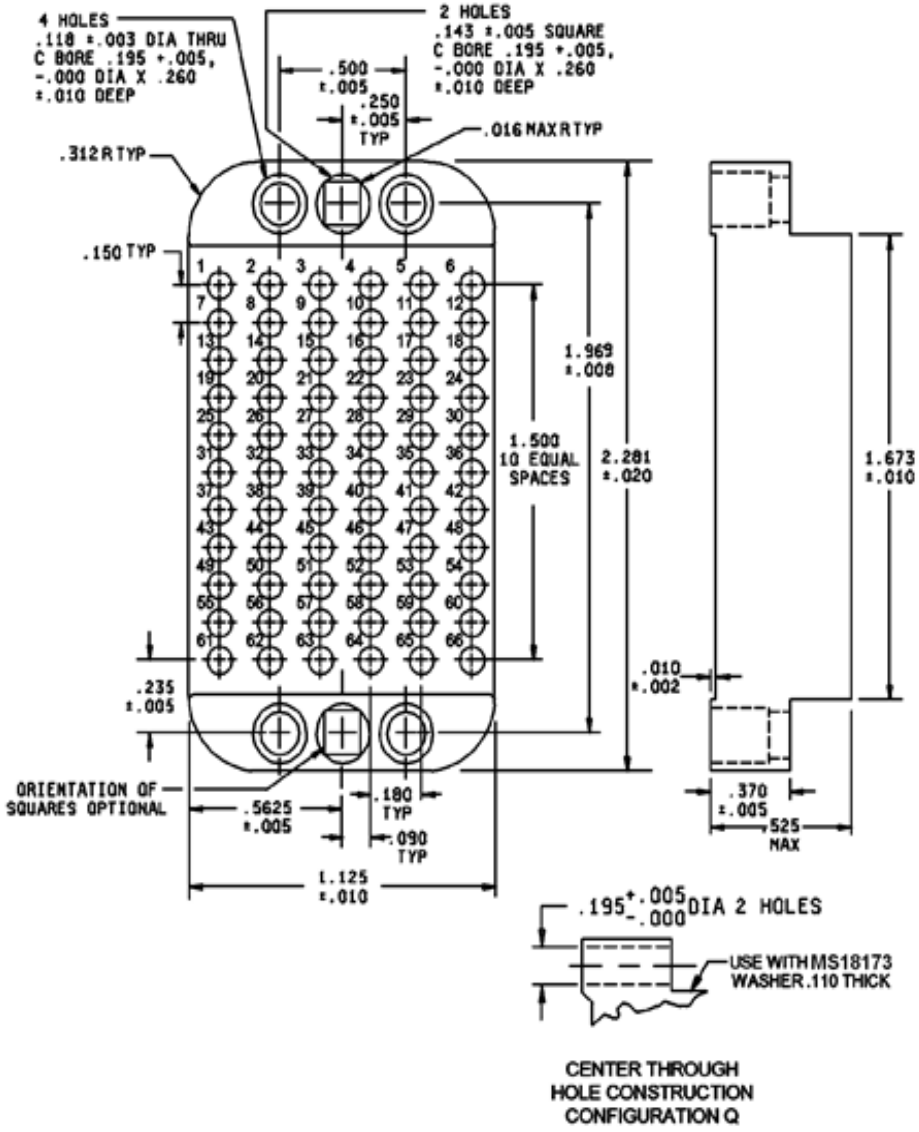
Configuration H - Suitable for all applications, except one-piece jackscrews.
Configuration P - For long jackscrews and shield only (through hole alternate construction).

Insert with 50 contacts

Insert designators H and P

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.020	0.51
.090	2.29
.110	2.79
.118	3.00
.143	3.63
.150	3.81
.180	4.57
.195	4.95
.235	5.97
.250	6.35
.260	6.60
.312	7.92
.370	9.40
.500	12.70
.525	13.34
.5625	14.288
1.125	28.58
1.500	38.10
1.673	42.49
1.969	50.01
2.281	57.94

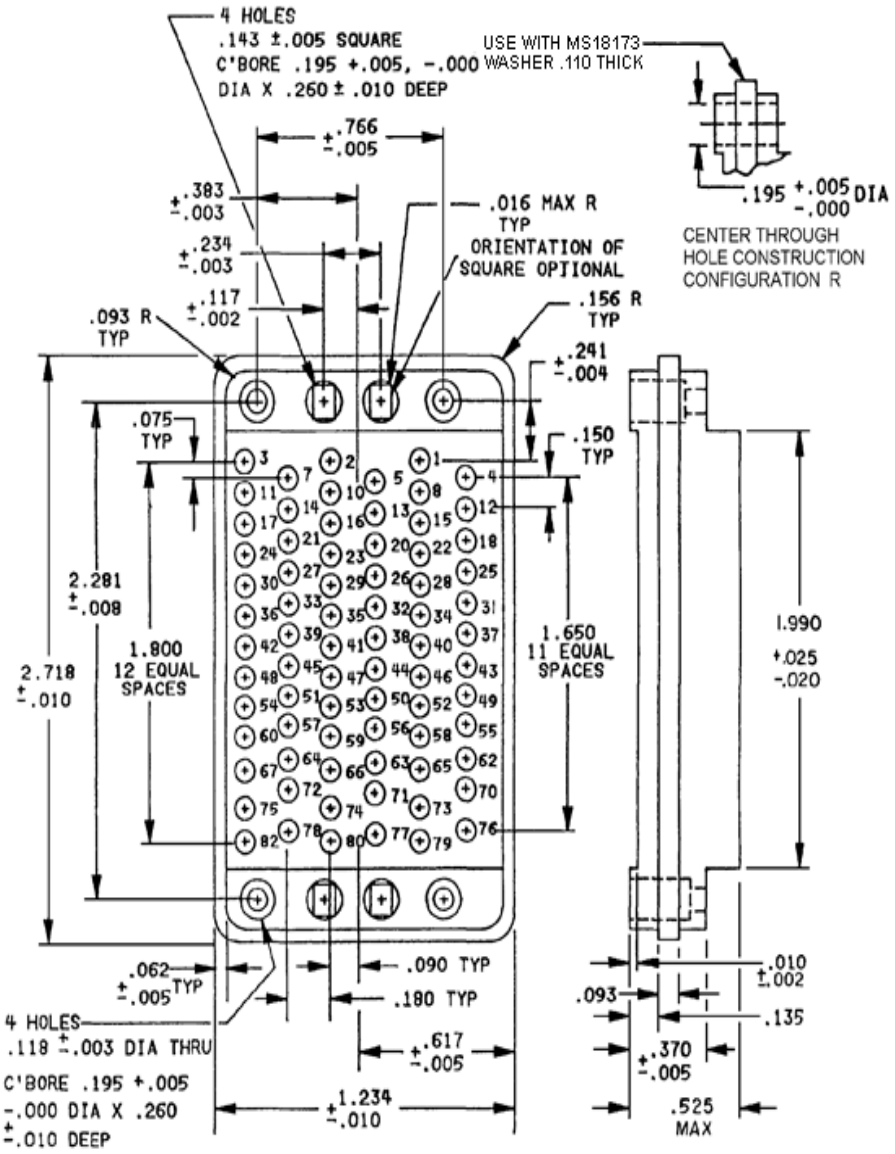
Configuration J - Suitable for all applications, except for one-piece jackscrews.
Configuration Q - For long jackscrews and shield only (through hole alternate construction).

Insert with 66 contacts

Insert designators J and Q

FIGURE 2. Insert arrangements - Continued.

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Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.004	0.10
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.020	0.51
.025	0.64
.062	1.57
.075	1.91
.090	2.29
.093	2.36
.110	2.79
.117	2.97
.118	3.00
.135	3.43
.143	3.63
.150	3.81
.156	3.96
.180	4.57
.195	4.95
.234	5.94
.241	6.12
.260	6.60
.370	9.40
.383	9.73
.525	13.34
.617	15.67
.766	19.46
1.234	31.34
1.650	41.91
1.800	45.72
1.990	50.55
2.281	57.94
2.718	69.04

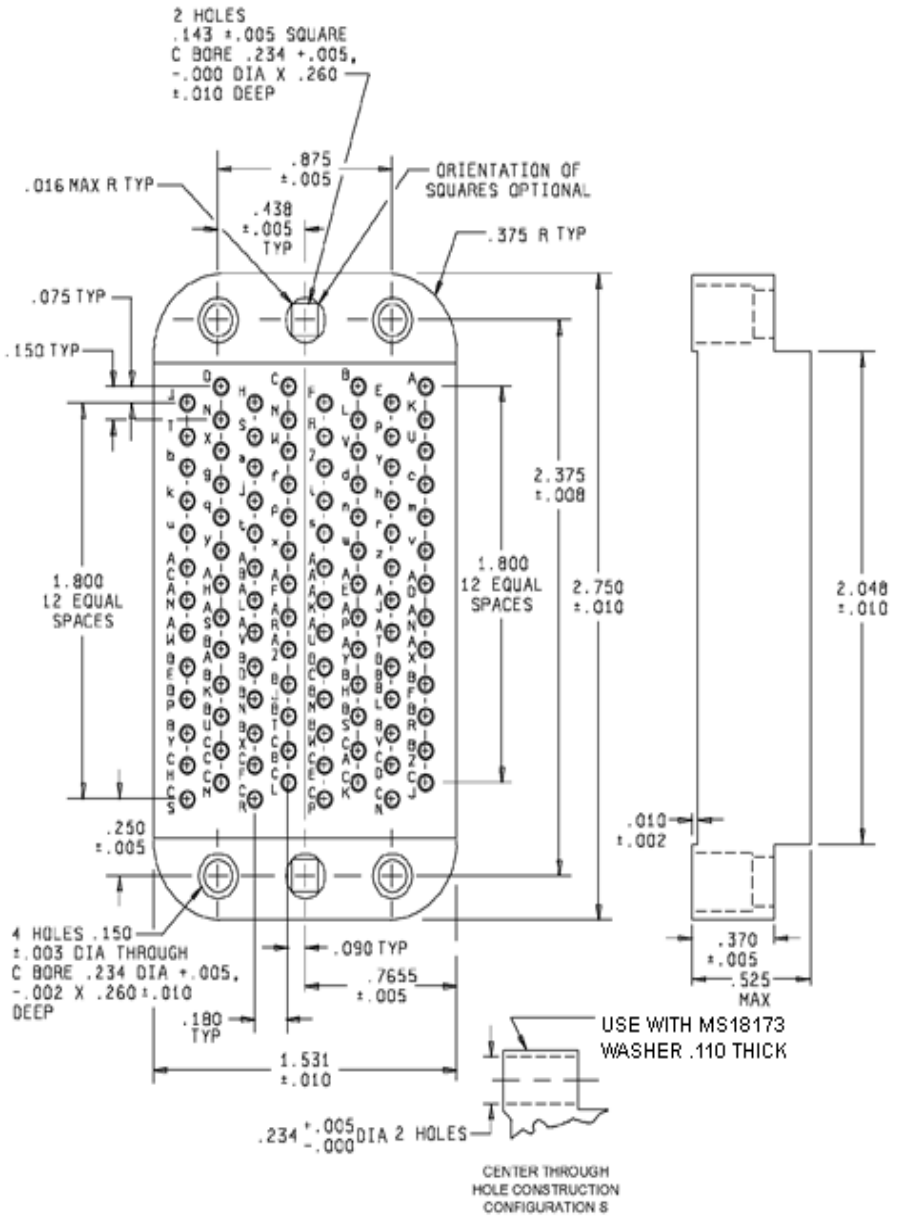
Configuration K - Suitable for all applications, except one-piece jackscrews.
Configuration R - For long jackscrews and shield only (through hole alternate construction).

Insert with 75 contacts

Insert designators K and R

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.075	1.91
.090	2.29
.110	2.79
.143	3.63
.150	3.81
.180	4.57
.234	5.94
.250	6.35
.260	6.60
.370	9.40
.375	9.53
.438	11.13
.525	13.34
.7655	19.44
.875	22.23
1.531	38.89
1.800	45.72
2.048	52.02
2.375	60.33
2.750	69.85

Configurations L - Suitable for all applications, except one-piece jackscrews.
Configurations S - For long jackscrews and shield only (through hole alternate construction).

Insert with 104 contacts

Insert designators L and S

FIGURE 2. Insert arrangements - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Insert material: See MIL-DTL-28748.

Mating connector: See MIL-DTL-28748/2 and MIL-DTL-28748/4.

Part or Identifying Number (PIN): As shown in the following example and table I: 1/

See table I for permissible PINs:

(See PIN codes)

M28748/3 –	A	1	H	L	1 A
Basic number of specification sheet	Insert designator	Shield and shield clamp location or retaining plate	Shell and polarization position	Jackscrews or guidepins	Contacts (see table II)

PIN codes:

Insert designator (See table III)	Number of contact positions
A	7
B	9
C	14
D	20
E	26
F	34
G	42
H	50
J	66
K	75
L	104
Insert designator through hole 5/	
M	34
N	42
P	50
Q	66
R	75
S	104

Shield and shield clamp location or retaining plate 2/

0 – No shield or retaining plate

Shield and shield clamp location

1 - top MS18192
2 - side MS18192
3 - top MS18193
4 - side MS18193

Retaining plate

5 – MS18198
6 – MS18199
7 – MS18200

Shell and polarization designators 3/

0 - No shell included

Plug shell

A – A of MS18203
B – B of MS18203
C – C of MS18203
D – D of MS18203
E – E of MS18203
F – F of MS18203
G – G of MS18203
H - Unpolarized MS18203

Receptacle shell

J – A of MS18204
K– B of MS18204
L– C of MS18204
M– D of MS18204
N– E of MS18204
P– F of MS18204
Q– G of MS18204
R– Unpolarized MS18204

Jackscrews or guidepins 4/

0 – No Jackscrews or guidepins.
F - Fixed jackscrews, MS18196 (not used with shield).
G - Guidepins, MS18197 (not used with shield).
L - Long jackscrews MS18194
S - Short jackscrews MS18195 (not used with shield).

Jack and guide location: The jackscrew female (turning or fixed) or guide socket shall be located by the A or 1 pin contact.

See footnotes on next page.

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1/ The number zero (0) is used to indicate which parts are not included.
2/ Retaining plate and shells are not available for insert size A.
3/ Polarization positions are for location of plug and receptacle shell pins and slots.
4/ Jackscrew designator L: Multi-piece jackscrews will be supplied with insert designators A through L. Long one-piece jackscrews will be supplied for insert designators M through S.
5/ Inserts M through S have through holes for use with long one-piece jackscrew designs only.
6/ Mold bodies and hardware from different manufacturers are not necessarily interchangeable.

TABLE I. Permissible PIN's (connector assembly).

<table>
<tr><td>Configuration</td><td colspan="5">Accessories</td></tr>
<tr><td>Receptacle</td><td>Insert</td><td>Shield and clamp position/ or retaining plate</td><td>Shell and polarization position</td><td>Jackscrews</td><td>Contacts</td></tr>
<tr><td rowspan="4">M28748/3-</td><td>A</td><td>0</td><td>0</td><td rowspan="4">0, F, G</td><td rowspan="13">See table II</td></tr>
<tr><td>B
C
D
E</td><td>0
Retaining Plate 5</td><td rowspan="3">0,
J thru R</td></tr>
<tr><td>F
G
H
K</td><td>0
Retaining Plate 6</td></tr>
<tr><td>J
L</td><td>0
Retaining Plate 7</td></tr>
<tr><td>Plug</td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="6">M28748/3-</td><td>A</td><td>0</td><td>0</td><td rowspan="6">0, L, G, S</td></tr>
<tr><td>B
C
D
E</td><td rowspan="2">0
Shield 1, 2</td><td rowspan="5">0,
A thru H</td></tr>
<tr><td>F
G
H
K</td></tr>
<tr><td>J
L</td><td>0
Shield 3, 4</td></tr>
<tr><td>M
N
P
R</td><td>0
Shield 1, 2</td></tr>
<tr><td>Q
S</td><td>0
Shield 3, 4</td></tr>
</table>

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TABLE II. Contact size percent 1/

Contact	Sizes 16-16 M39029/34-273 (percent)	Sizes 16-20 M39029/34-272 (percent)	Sizes 20-20 M39029/34-271 (percent)
1A	100	0	0
1B	90	0	0
1C	80	0	0
1D	70	0	0
1E	60	0	0
1F	50	0	0
1G	40	0	0
1H	30	0	0
1J	20	0	0
1K	10	0	0
1L	0	0	0
2A	0	100	0
2B	0	90	0
2C	0	80	0
2D	0	70	0
2E	0	60	0
2F	0	50	0
2G	0	40	0
2H	0	30	0
2J	0	20	0
2K	0	10	0
3A	0	0	100
3B	0	0	90
3C	0	0	80
3D	0	0	70
3E	0	0	60
3F	0	0	50
3G	0	0	40
3H	0	0	30
3J	0	0	20
3K	0	0	10

1/ Contact percent size = number of contact positions times the percent in table II. Example: 34 X .80 = 27.2 or 27 contacts supplied. Values of .5 and above are rounded to the next contact.

Crimp tool M22520/1-01.
Positioner M22520/1-03 (positioner A color blue/red).
Installation tool M81969/18-01.
Removal tool M81969/20-01.

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For superseded insert Military Standard sheet PIN's to M28748 PIN's see table III.

TABLE III. Supersession data (inserts).

MS PIN	M28748/3 PIN	Number of contact positions
MS14010-1	M28748/3-A0001L	7
MS14007-1	M28748/3-B0001L	9
MS18174-1	M28748/3-C0001L	14
MS18176-1	M28748/3-D0001L	20
MS14008-1	M28748/3-E0001L	26
MS18178-1	M28748/3-F0001L	34
MS18180-1	M28748/3-G0001L	42
MS18182-1	M28748/3-H0001L	50
MS18184-1	M28748/3-J0001L	66
MS18187-1	M28748/3-K0001L	75
MS18189-1	M28748/3-L0001L	104
MS18178-2	M28748/3-M0001L	34 1/
MS18180-2	M28748/3-N0001L	42 1/
MS18182-2	M28748/3-P0001L	50 1/
MS18184-2	M28748/3-Q0001L	66 1/
MS18187-2	M28748/3-R0001L	75 1/
MS18189-2	M28748/3-S0001L	104 1/

1/ Insert through hole designator. Inserts –2 (M through S) have through holes for use with long one-piece jackscrew designs only.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Reference documents. In addition to MIL-DTL-28748, this document references the following:

MIL-DTL-28748/2
MIL-DTL-28748/4
MS18173
MS18192
MS18193
MS18194
MS18195
MS18196
MS18197
MS18198
MS18199
MS18200
MS18203
MS18204

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2017-002)

Review activities:
Army - AT, MI
Navy - AS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.